

34 The Origin and Evolution of Vertebrates

KEY CONCEPTS

- 34.1** Chordates have a notochord and a dorsal, hollow nerve cord p. 719
- 34.2** Vertebrates are chordates that have a backbone p. 722
- 34.3** Gnathostomes are vertebrates that have jaws p. 725
- 34.4** Tetrapods are gnathostomes that have limbs p. 730
- 34.5** Amniotes are tetrapods that have a terrestrially adapted egg p. 734
- 34.6** Mammals are amniotes that have hair and produce milk p. 741
- 34.7** Humans are mammals that have a large brain and bipedal locomotion p. 748

Study Tip

Make a table: Humans are mammals—and we are also amniotes, tetrapods, lobe-fins, osteichthyans, gnathostomes, vertebrates, and chordates (see Figure 34.2). As you read the chapter, list derived characters for these clades and describe whether these traits have been retained, modified, or lost in the human lineage.

Clade	Derived characters	Found in humans?
Chordates	Notochord	Modified (gelatinous disks between vertebrae)
	Post-anal tail	Lost

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For Students (in eText and Study Area)

- Get Ready for Chapter 34
- Video: Sea Horse Camouflage
- HHMI Video: Great Transitions: The Origins of Humans

For Instructors to Assign (in Item Library)

- Tutorial: Chordates
- Problem-Solving Exercise: Declining Amphibian Populations



Figure 34.1 Small and slender, this inconspicuous species (*Myllokunmingia fengjiaoa*) lived in the ocean 530 million years ago, along with an immense variety of dangerous and well-defended invertebrate animals. Although lacking claws or armor, this ancient species was closely related to one of the most successful groups of animals on Earth: the vertebrates, animals with a backbone.

What are some key characteristics that have appeared during vertebrate evolution?

